Breast cancer classification on EMBED

A deep-learning study on the Emory Breast Imaging Dataset

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Scope. Final report on the EMBED breast-imaging run, with embedded figures and proof-of-work log panels. The mammogram and Grad-CAM images in this document are from the open EMBED release.

1. Summary

We trained and evaluated breast-image classifiers on EMBED (Emory Breast Imaging Dataset [1], S3 `embed-dataset-open`, us-east-1) for two tasks:

- **Task A — malignant vs. benign** on exams within 1–180 days of a biopsy (3,407 images, 1,757 breast exams).
- **Task B — BI-RADS category** (worst `asses` per breast exam) on fold 0 (22,086 images, 11,268 breast exams, each class capped at 2,500 exams).

The best Task A model on the held-out test set (348 exams, 112 cancers) averages two image runs of ResNet-50, EfficientNetV2-S and ConvNeXt-Tiny (768×512 and 1,536×1,024, six models in all) and adds exam type, age and the radiologist's pre-biopsy findings and assessment:

- **Test AUC-ROC = 0.826 (95% CI 0.779–0.872).**
- Accuracy 78.7%, sensitivity 57%, specificity 89%, NPV 81%.
- On the 20% of exams (71 of 348) where the model is at least 90% confident, 94% of predictions are correct (67 of 71); of 54 confident "benign" calls, 2 were cancers.

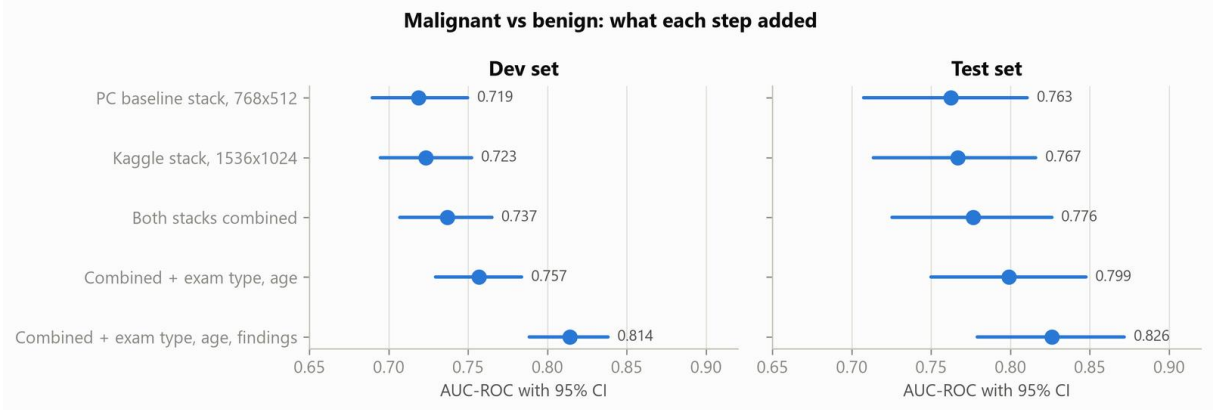


Figure 1. *AUC on Task A at each step, on dev and test: 768×512 stack, 1,536×1,024 stack, both combined, then exam type and age, then the radiologist's findings. Lines show the 95% bootstrap CI.*

The overall accuracy of 78.7% does not meet the 90% accuracy target the team set at the start of the project. The model is not fit for clinical use. The report documents how the result was produced and lists the limitations.

A short, honest headline: on biopsy-indicated exams at one institution, the best image+findings model separates cancer from benign at roughly AUC 0.83. Much of that comes from the radiologist's own reading: on dev, the reading with exam type and age, without images, reaches AUC 0.797 against 0.809 with images (see §5). It is not a screening model, and it has not been validated outside EMBED.

2. Data

Source. EMBED (Emory Breast Imaging Dataset) [1], open release on S3 (20% of the full 2D and C-view dataset, research use under Emory's data use agreement), bucket `embed-dataset-open` in `us-east-1`.

2.1 Task A — malignant vs. benign

- Label from `path_severity`: *malignant* if any result on the breast is 0 or 1 (invasive or in-situ cancer); exam excluded if any is 2, 3 or 5 (high-risk, borderline or non-breast malignancy); *benign* otherwise.
- Only exams performed 1 to 180 days before a biopsy are included, so every exam in Task A led to a biopsy.
- Final sizes: **3,407 images, 1,757 breast exams → dev 1,409 (457 cancers, prevalence 32.4%), test 348 (112 cancers, prevalence 32.2%)**. Each breast in an exam counts separately; dev comes from 1,210 patients and test from 303.
- Patient-level split (no patient appears in both dev and test). The test set was locked at the start and opened once at the end, after the decisions in `test_plan_malignant.md` were written.



Figure 2. *Task A cohort: age at exam, breast density and scanner manufacturer.*

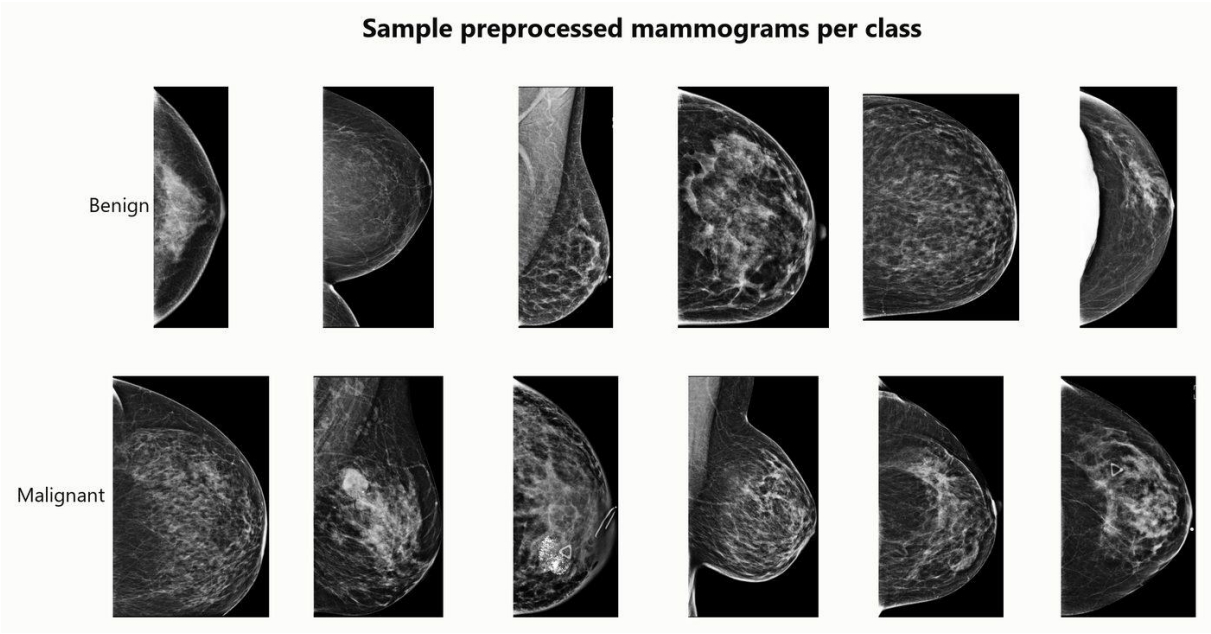


Figure 3. *Example EMBED images from Task A after crop and resize (dataset-provided imagery).*

2.2 Task B — BI-RADS

- Label is the worst asses for the breast exam (categories 0, 1, 2, 3, 4, 5).
- Only fold 0 of the project's patient-level split is used (budget-driven decision). Each BI-RADS class is capped at 2,500 breast exams by random sampling to limit image downloads, so the class mix is not the natural one.
- Final sizes: **22,086 images, 11,268 breast exams → dev 9,014 (fold 0 validation 1,826), test 2,254**.

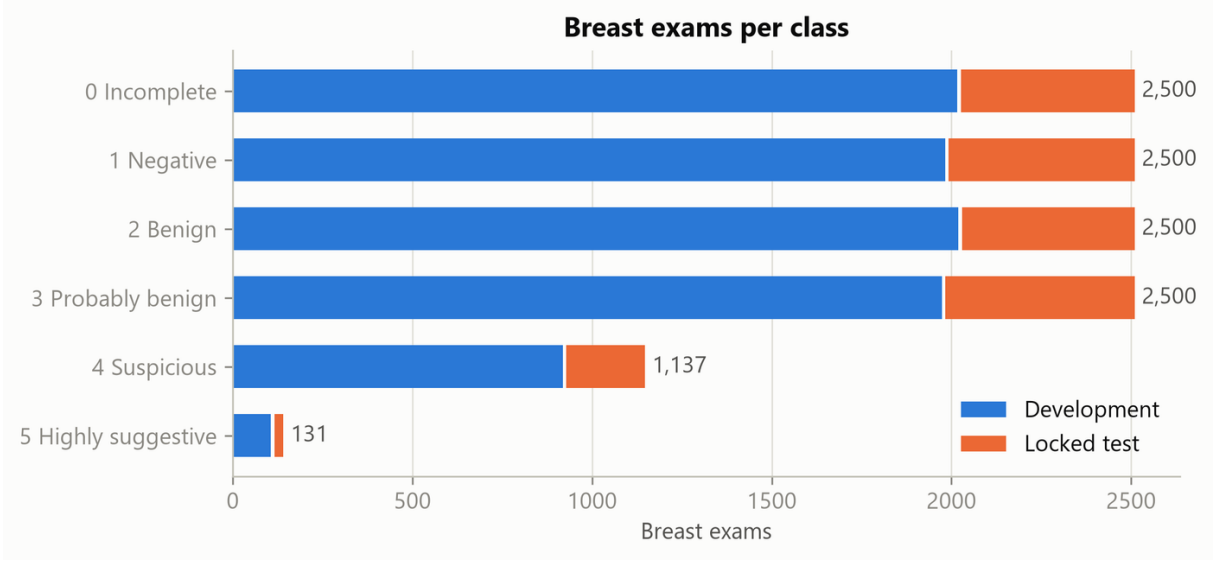


Figure 4. *Task B cohort: breast exams per BI-RADS class in dev and the locked test set, after the cap.*

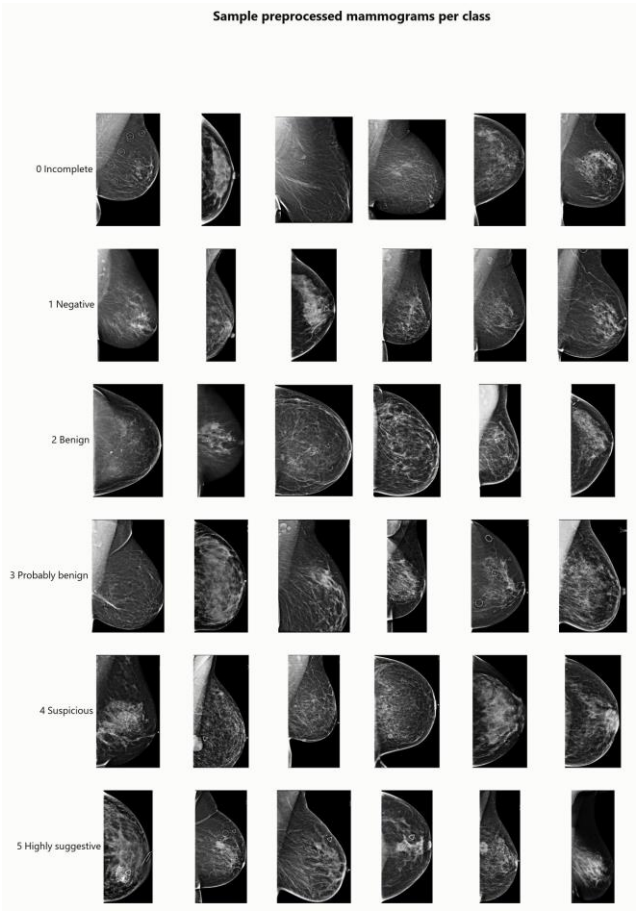


Figure 5. *Example EMBED images for Task B across BI-RADS categories.*

2.3 Processing and the crop fix

Images are cropped to the breast ROI, flipped so the breast always sits on the same side, resized and converted to PNG. A crop regression discovered during the first pass (tissue cut in 4 of 40 sampled images) was fixed in the v2 pipeline; it is documented in `docs/fix_log.md`.

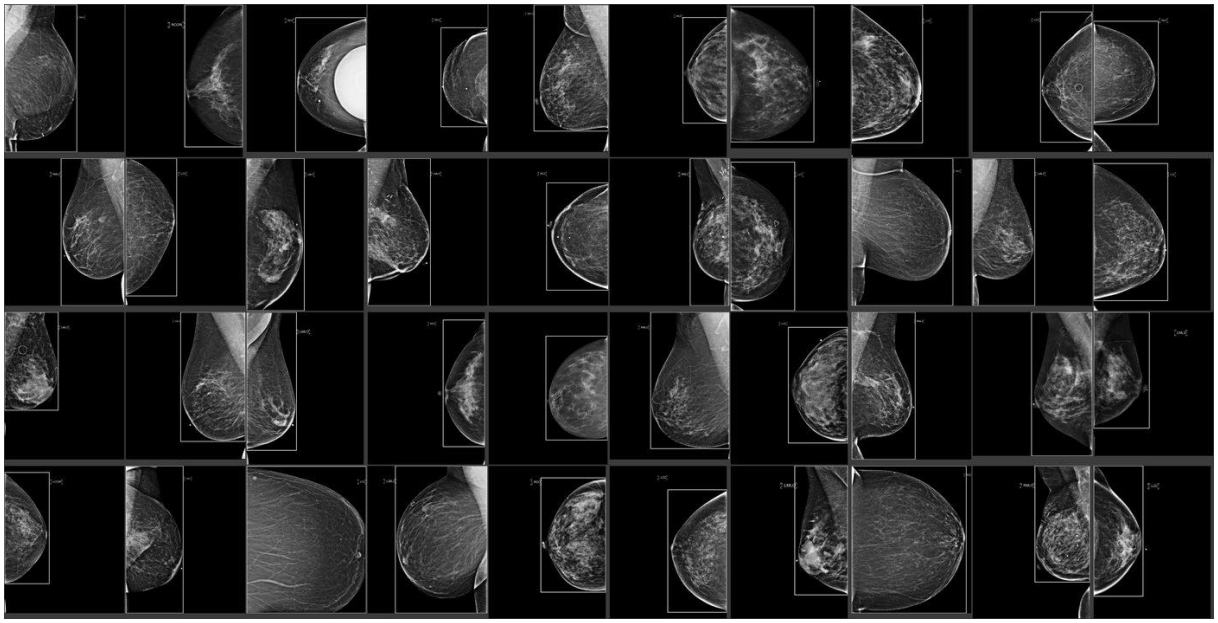


Figure 6. *Crop check: side-by-side of the old crop (left) and the fixed crop (right) on sample images. The old crop occasionally sliced through tissue; the fix is the one used in all v2 results.*

2.4 Non-image inputs

- `stack_clinical`: exam type (screening or diagnostic, from the exam description) and patient age.
- `stack_reading`: additionally includes what the radiologist recorded before the biopsy (Task A only): mass shape, margin and density, calcification findings and distribution, other findings, and the BI-RADS assessment. Pathology, grade, stage, receptors and surgery are the answer and are **never** used as inputs.

3. Methods

Models. Three image backbones: **ResNet-50** [2], **EfficientNetV2-S** [3], and **ConvNeXt-Tiny** [4]. Each is trained at two resolutions:

- `outputs/` — PC baseline, 768×512, old crop, Task A only. It is part of the combined Task A result (see §5).
- `outputs_v2/` — Kaggle, **1,536×1,024** for Task A and **1,024×768** for Task B, corrected crop. All Task B results come from this run.

Training protocol.

- Task A: 5 dev folds, patient-level cross-validation. Task B: fold 0 only.
- Cross-entropy loss with class-weighted sampling for both tasks (label smoothing 0.05 on Task B). Training augmentations: vertical flip, rotation up to ±10°, scale 0.9–1.1, shift up to ±5%, brightness and contrast changes.
- Stop-and-fix policy on any run-time error: a failing run is diagnosed and re-launched, never silently retried.
- Checkpoints: `best.pt` is kept on the PC; `last.pt` for 14 of the 15 Task A Kaggle runs remains only on Kaggle (`kaggle/fetch_output.py --all`).

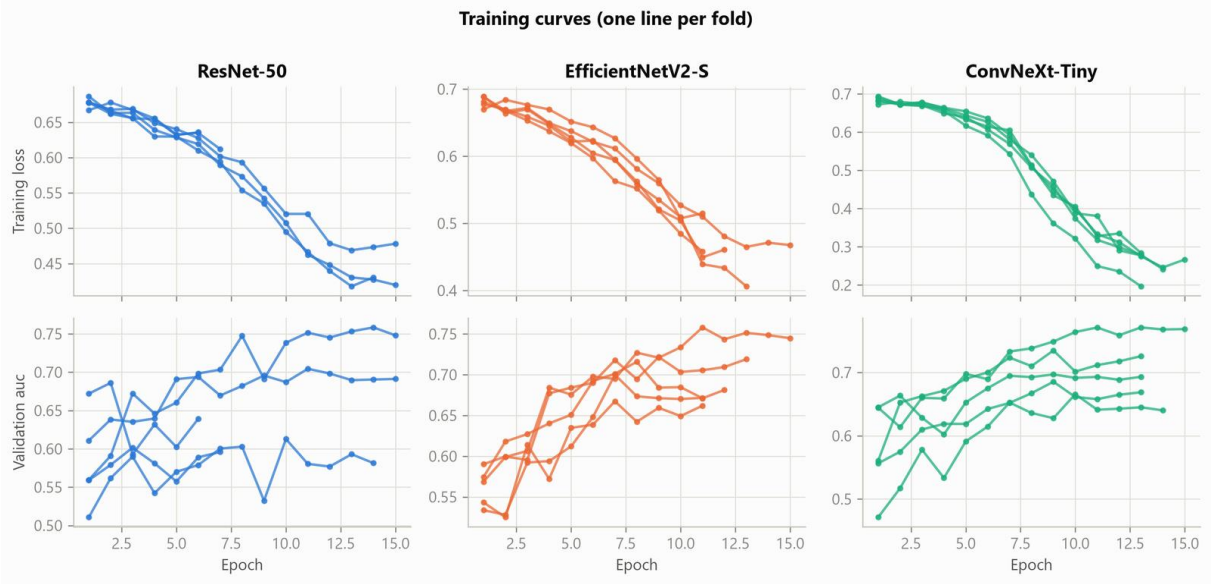


Figure 7. *Training curves for Task A: training loss and validation AUC per fold, three backbones.*

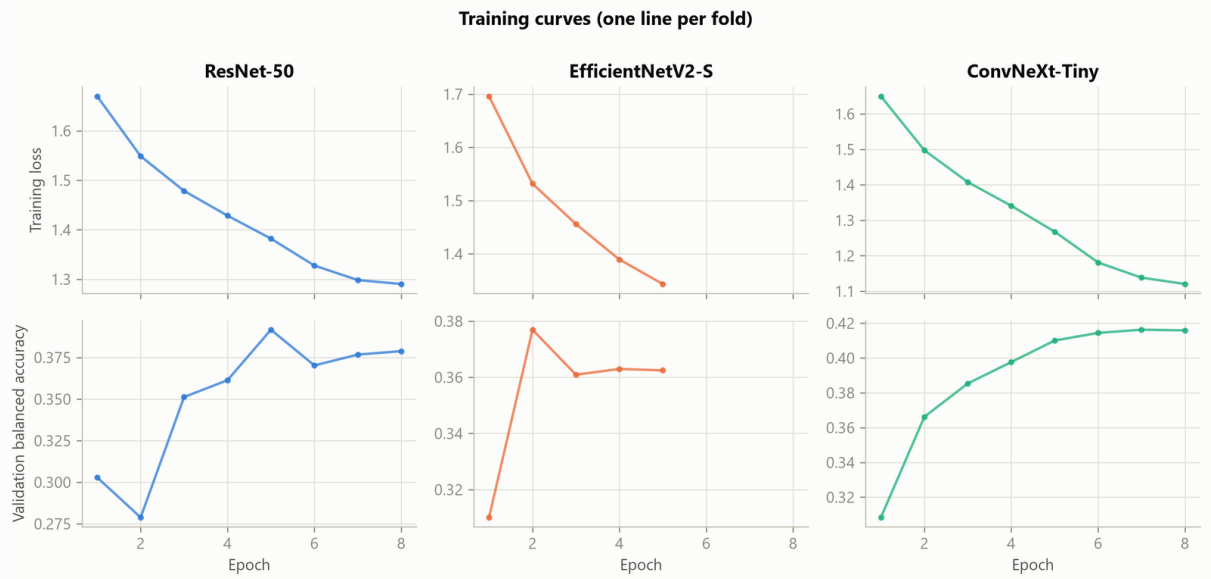


Figure 8. *Training curves for Task B (BI-RADS): training loss and validation balanced accuracy, fold 0 only.*

Combining models. Within each run, the three image models were combined by the method that scored best on dev among a plain mean, a logistic stacker and LightGBM: the mean for Task A, the logistic stacker for Task B. Exam type, age and the radiologist's findings were added through a logistic model on the model outputs plus those inputs, evaluated out-of-fold on dev. The final Task A result averages, in log-odds, the outputs of the 768×512 and 1,536×1,024 runs.

Evaluation.

- All metrics are computed at the **breast-exam** level (per-image predictions are averaged to the exam level, mirroring how a radiologist reads a case).
- **Patient-level bootstrap** is used for 95% CIs on AUCs and operating-point metrics (2,000 resamples for the combined Task A results, 1,000 within each run); a patient and all of their exams are resampled together.
- Operating thresholds for Task A are chosen on dev (default 0.5, Youden's J, max-accuracy, and the threshold achieving sensitivity ≥ 0.90) and then **frozen** before touching the test set. Result tables use the max-accuracy threshold.

Reproducibility. `outputs_v2/proof/manifest.csv` records the SHA-256 of 336 files (checkpoints, predictions, cohort tables, metrics); `outputs_v2/proof/kaggle_env.json` and `kaggle_pip_freeze.txt` record the Kaggle environment; the Task A test plan, written before the test set was opened, is in `outputs_v2/proof/test_plan_malignant.md`.

4. Results

Every number in the tables below is read from `outputs_v2/metrics/final_summary.csv` and `outputs_v2/metrics/a_combined.json`.

4.1 Task A — malignant vs. benign (test, 348 exams, 112 cancers)

Always-predict-benign baseline accuracy = 67.8%. The combined six-model system, with thresholds frozen on dev (max-accuracy threshold):

Inputs	AUC-ROC (95% CI)	Accuracy	Sensitivity	Specificity	PPV	NPV	Dev AUC
Images only	0.776 (0.725–0.826)	73.3%	25.9%	95.8%	74.4%	73.1%	0.737
+ exam type, age	0.799 (0.750–0.848)	73.6%	40.2%	89.4%	64.3%	75.9%	0.757
+ radiologist findings	0.826 (0.779–0.872)	78.7%	57.1%	89.0%	71.1%	81.4%	0.814

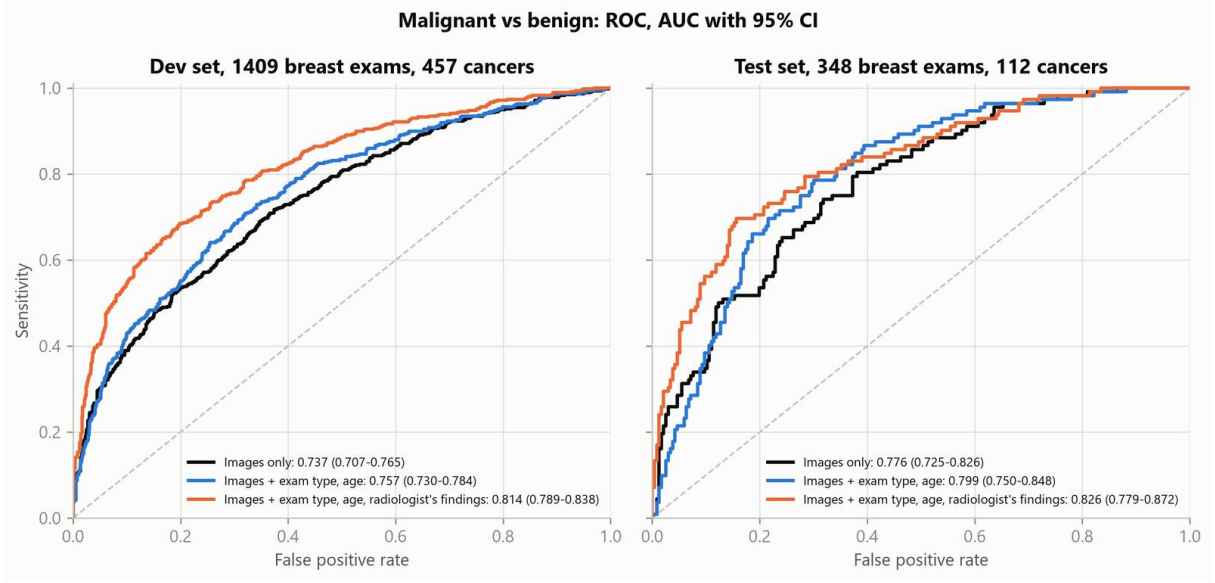


Figure 9. *ROC curves on the Task A dev and test sets for the three combinations.*

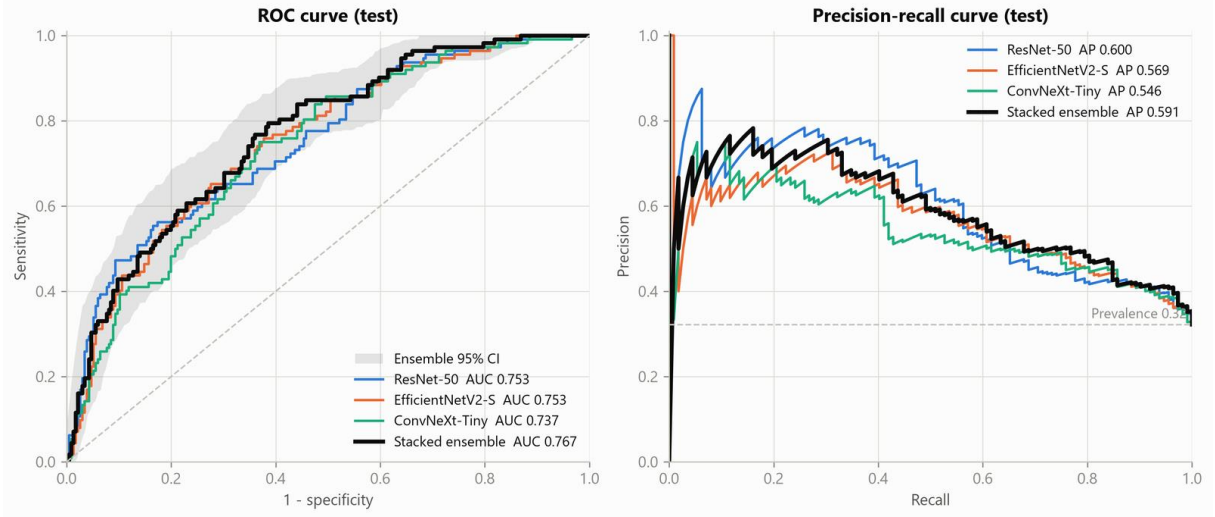


Figure 10. *ROC and precision-recall curves on test for the three models of the 1,536×1,024 run and their mean (labelled "stacked ensemble").*

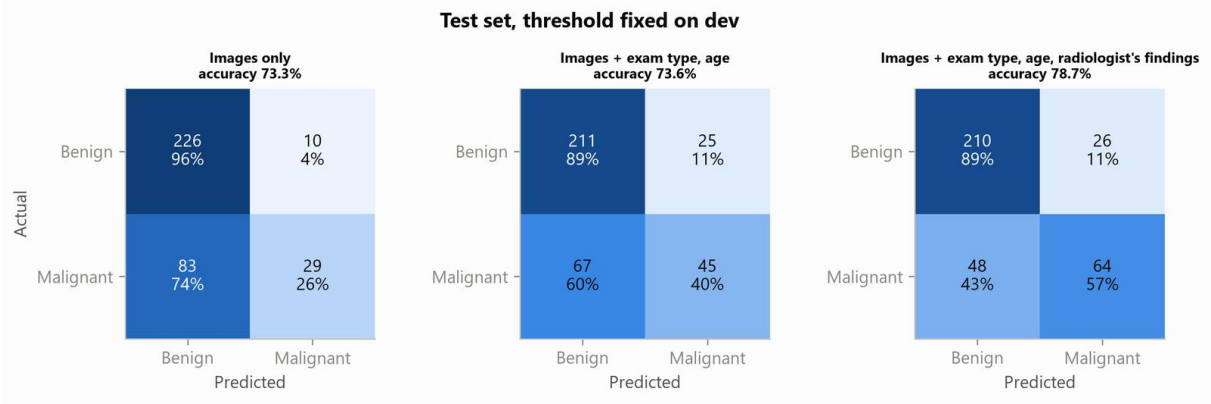


Figure 11. *Confusion matrices on test at the max-accuracy threshold, for all three combinations.*

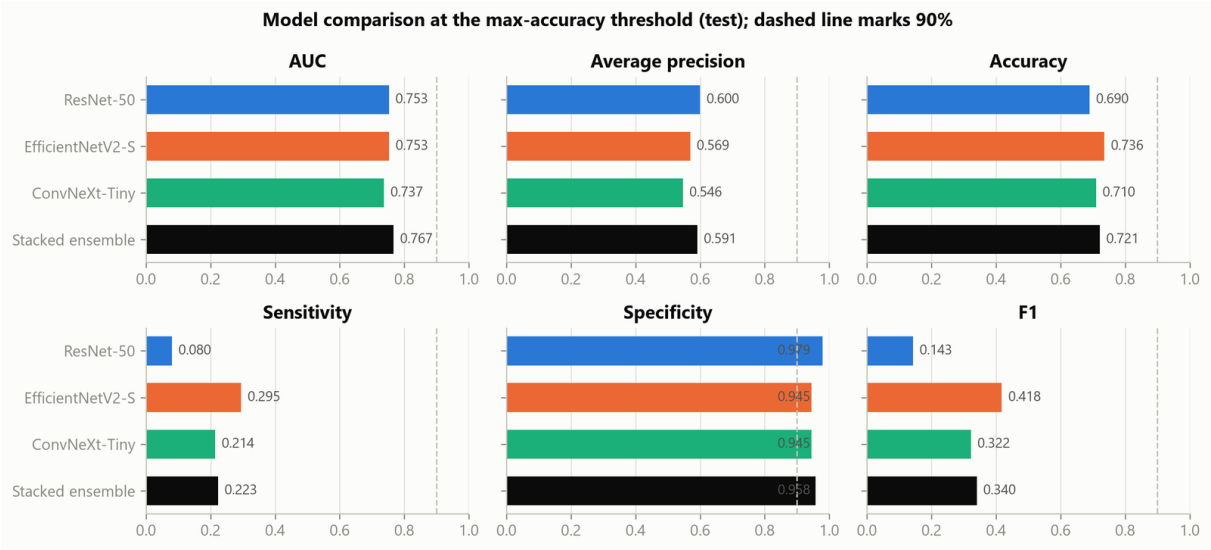


Figure 12. *Per-model comparison on test at the max-accuracy threshold, 1,536×1,024 run: six metrics for each backbone and their mean (labelled "stacked ensemble").*

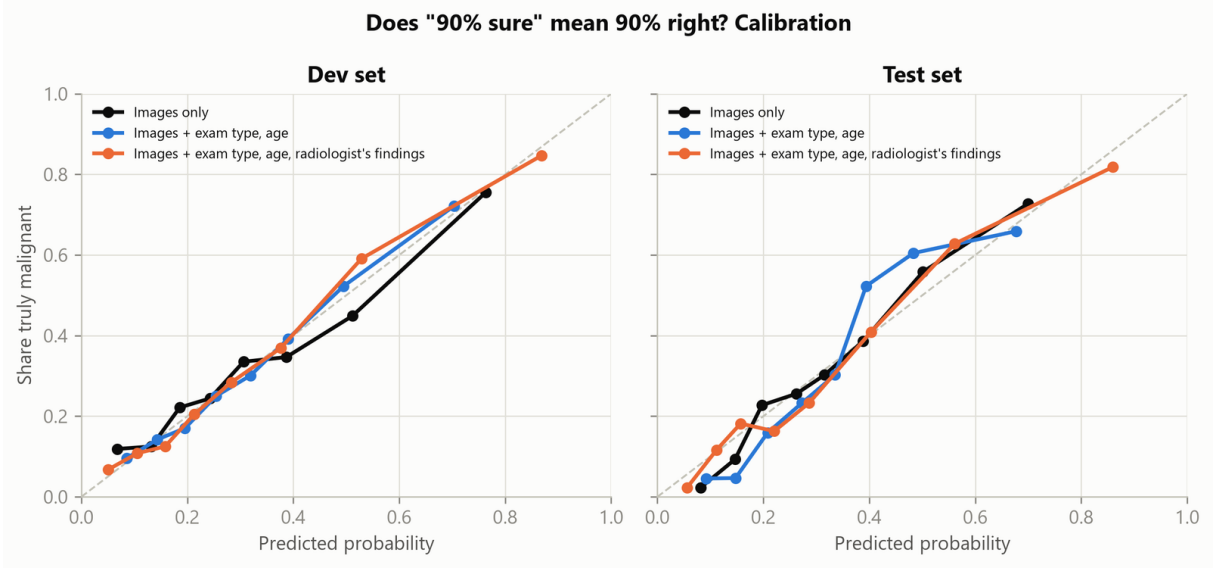


Figure 13. *Calibration (reliability) on dev and test for the three combinations. Test ECE: images only 0.054, + exam type and age 0.063, + findings 0.044.*

Confidence vs. coverage. Keeping only exams where the model's predicted probability is at least 0.9 or at most 0.1 ($|p - 0.5| \geq 0.4$):

Inputs	Coverage (n)	Overall correct	Benign calls correct	Cancers missed as benign
Images only	10.1% (35)	94.3%	100% (32/32)	0 of 32
+ exam type, age	7.5% (26)	100% (26/26)	100% (25/25)	0 of 25
+ findings	20.4% (71)	94.4%	96.3% (52/54)	2 of 54

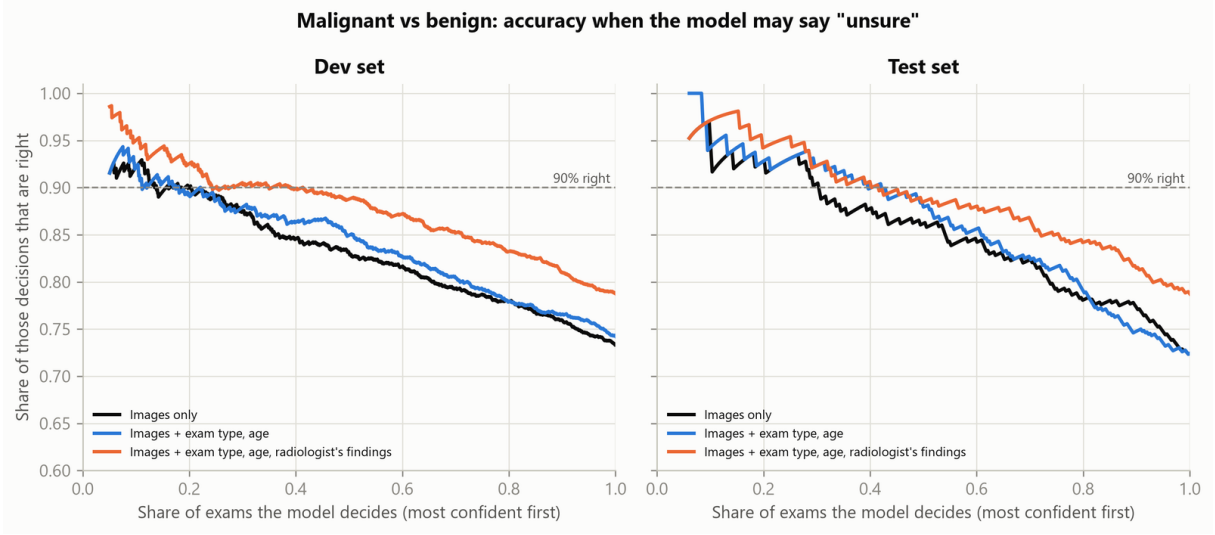


Figure 14. *Coverage vs. accuracy on dev and test: the fewer exams the model decides (most confident first), the more of those decisions are right.*

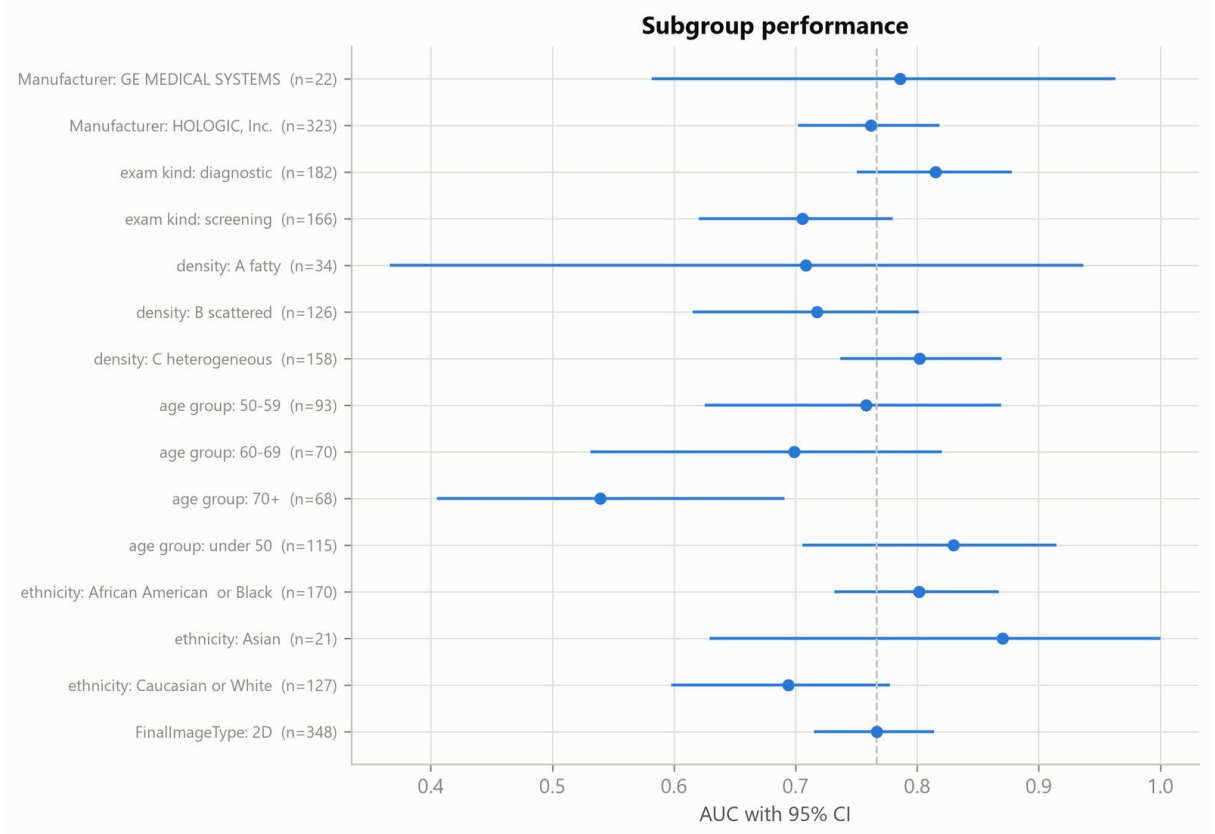


Figure 15. *Task A subgroup AUC on test for the image-only 1,536×1,024 run: scanner, exam kind, density, age and ethnicity.*

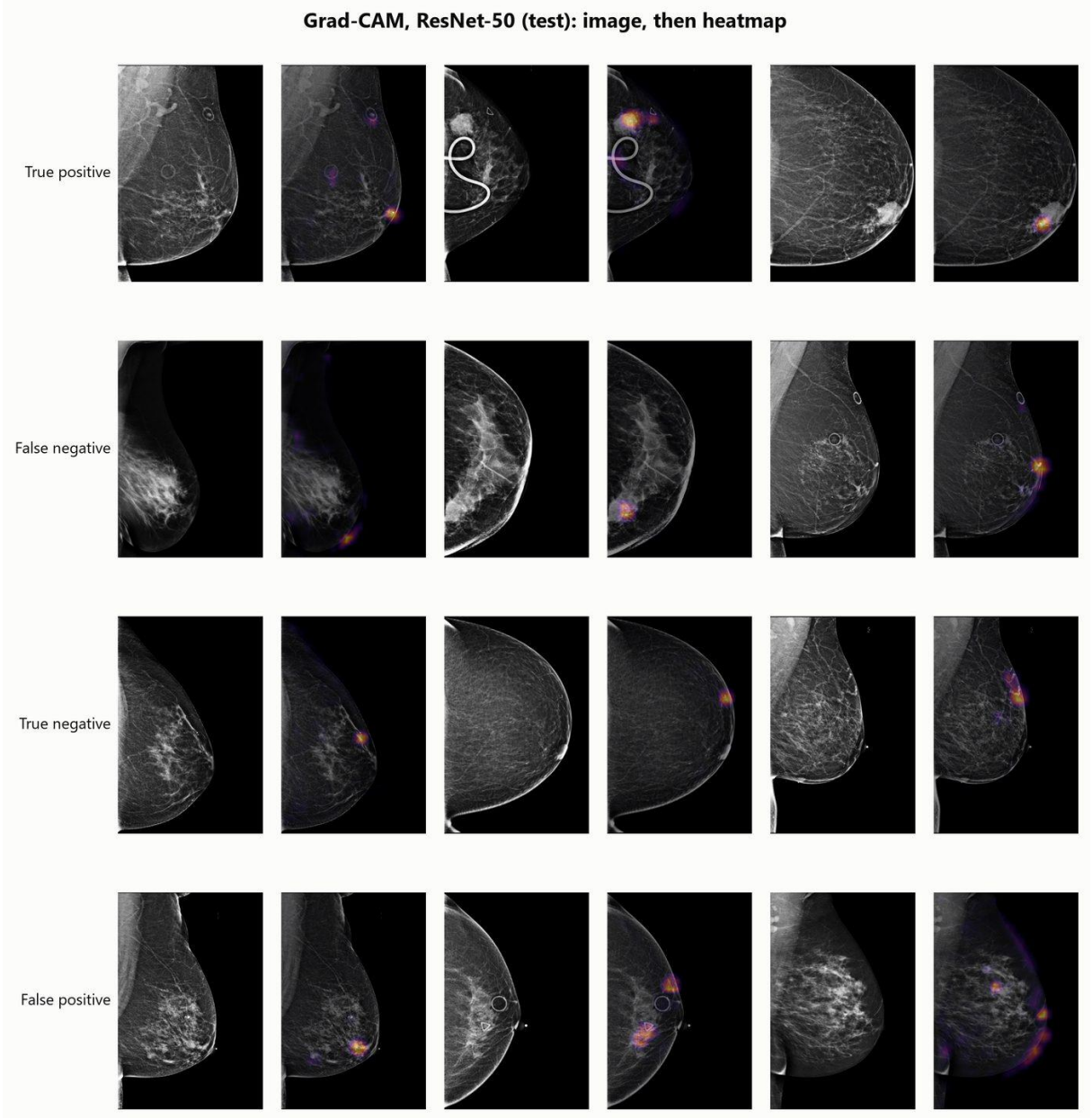


Figure 16. *Grad-CAM [5] maps on Task A test exams. Attention is mostly on tissue but with some weight on image markers — see §4.3.*

Individual model stacks and resolution. The 1,536×1,024 Kaggle stack (image-only) reaches dev AUC 0.723 and test AUC 0.767; the 768×512 baseline stack reaches 0.719 and 0.763. Doubling the resolution gave at most ~0.004 AUC; the gains documented above come from combining the two runs and from adding the clinical and reading features, not from the finer images.

Subgroups. Figure 15 uses the image-only 1,536×1,024 run (test AUC 0.767). Groups are small and intervals wide. AUC is 0.539 (95% CI 0.405–0.691, n = 68) for age 70+, where the interval includes chance, against 0.830 (0.705–0.914, n = 115) under age 50. It is 0.815 (0.750–0.878, n = 182) on

diagnostic exams and 0.706 (0.620–0.780, n = 166) on screening exams. These differences are descriptive only.

4.2 Task B — BI-RADS (test, 2,254 exams)

Inputs	Accuracy	Balanced accuracy	Macro AUC
Images only (stack of 3 models)	43.6%	39.1%	0.781
+ exam type, age	54.7%	47.8%	0.860

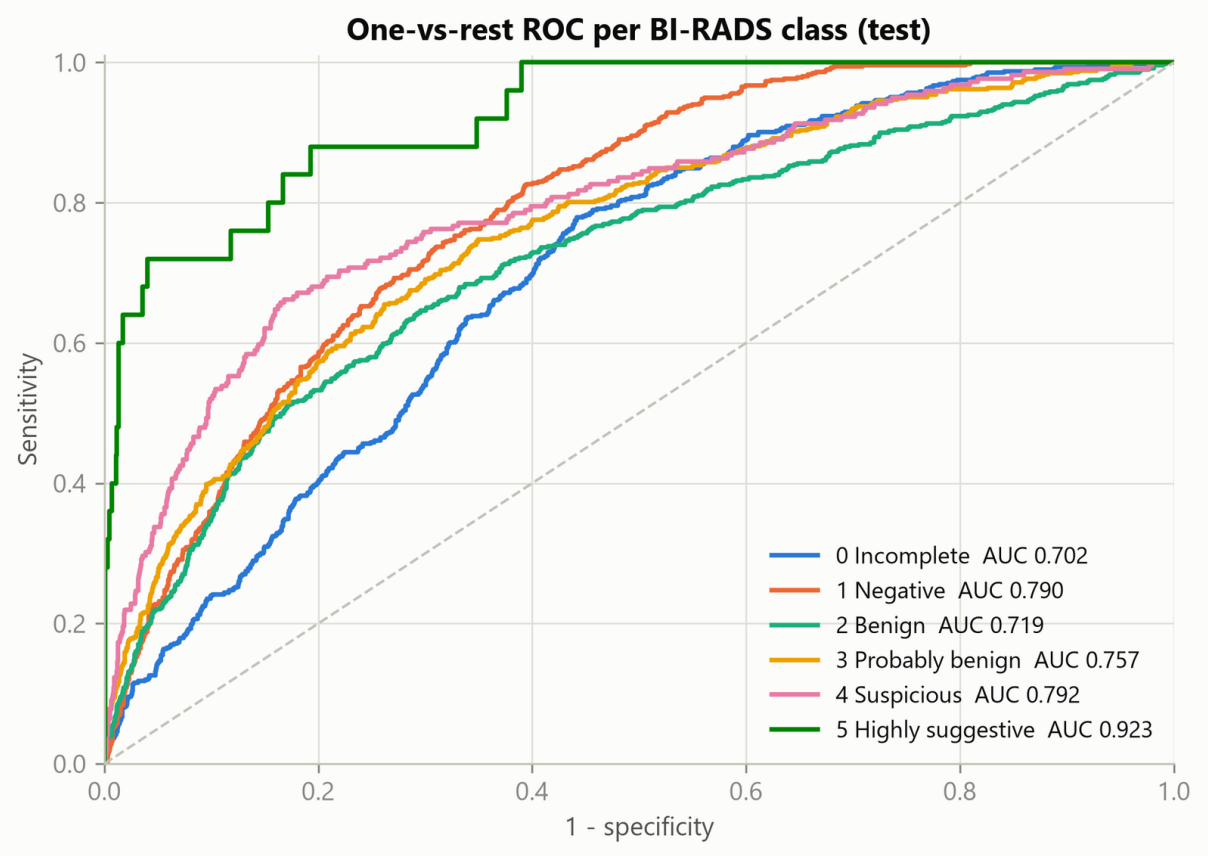


Figure 17. *One-vs-rest ROC per BI-RADS class on test.*

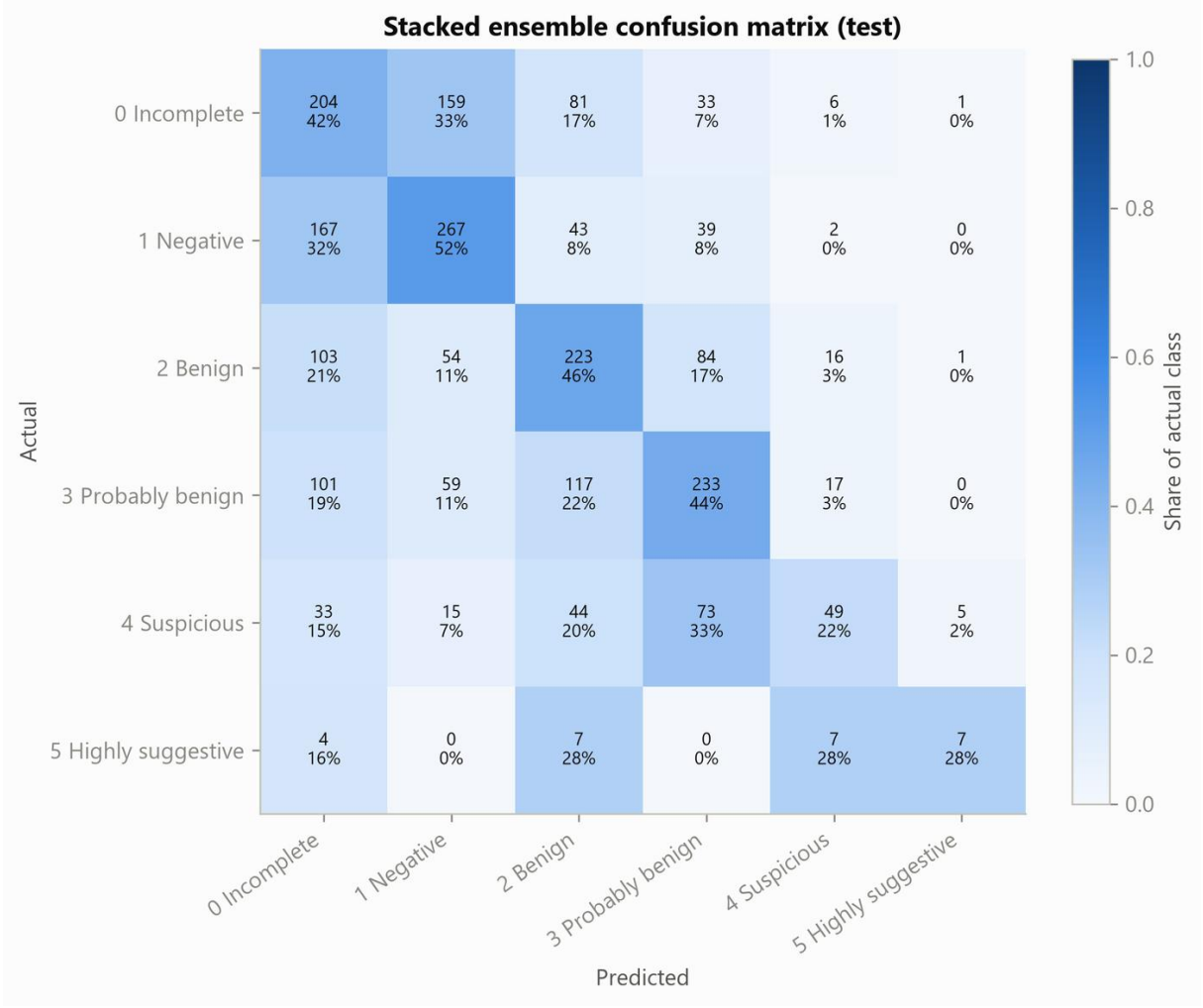


Figure 18. *BI-RADS confusion matrix on test.*

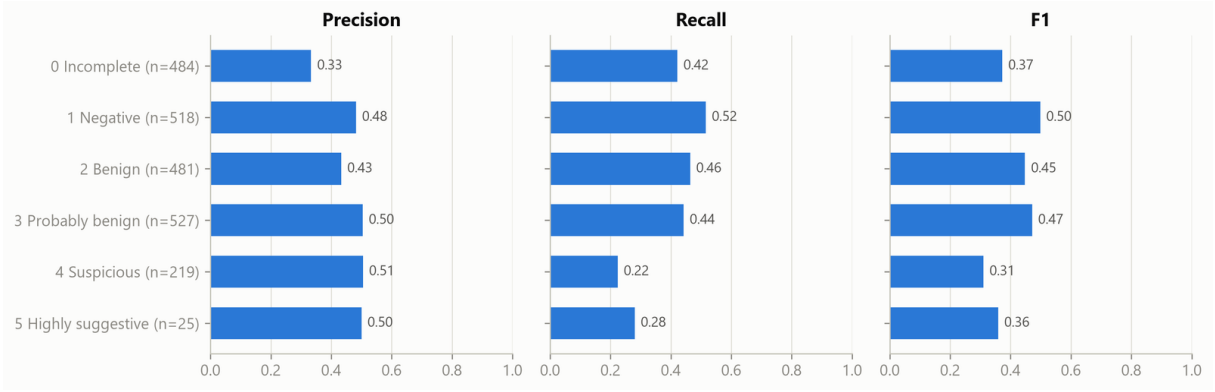


Figure 19. *Per-class precision, recall and F1 on test.*

The stack of image models **did not beat ConvNeXt-Tiny on its own** (p = 0.67). Exam type alone, with no image at all, reaches 42.8% dev accuracy and macro AUC 0.735 (`b_metadata_baseline.json`); the image stack reaches 42.7% and 0.759. This suggests that **much**

of the apparent BI-RADS "image skill" may reflect exam type, which the images can reveal through markers. Markers were not tested on Task B, so this is not proven:

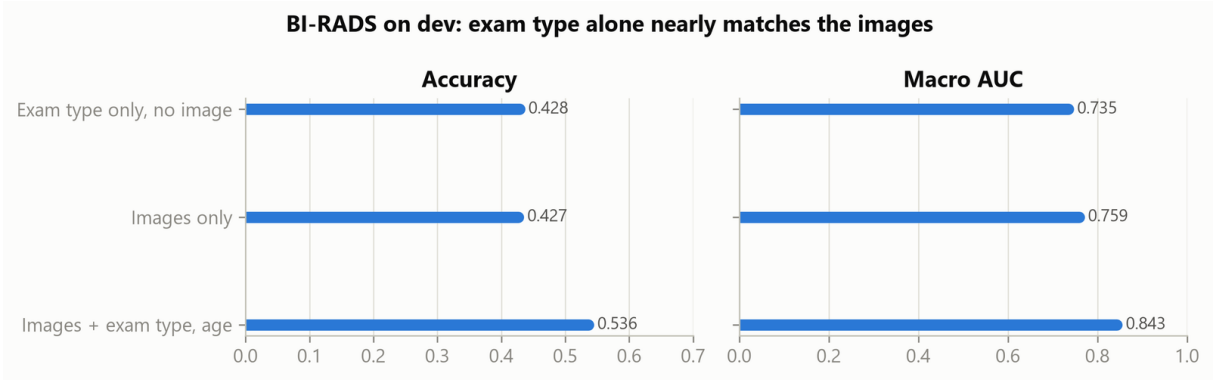


Figure 20. *BI-RADS on dev: exam type alone nearly matches the image stack, which suggests the image-only model may be learning the exam-type cue rather than lesion content.*

4.3 Marker-reliance test

Rings, chips and other overlays on EMBED mammograms can leak information about the exam type (screening vs. diagnostic) and about the laterality. We retrained ConvNeXt-Tiny on Task A with a simple marker detector and inpainting, and compared it with the baseline run:

- Dev AUC baseline = 0.6975
- Dev AUC with markers removed = 0.6881
- Δ = −0.009 (95% CI −0.037 to +0.017), n = 1,409

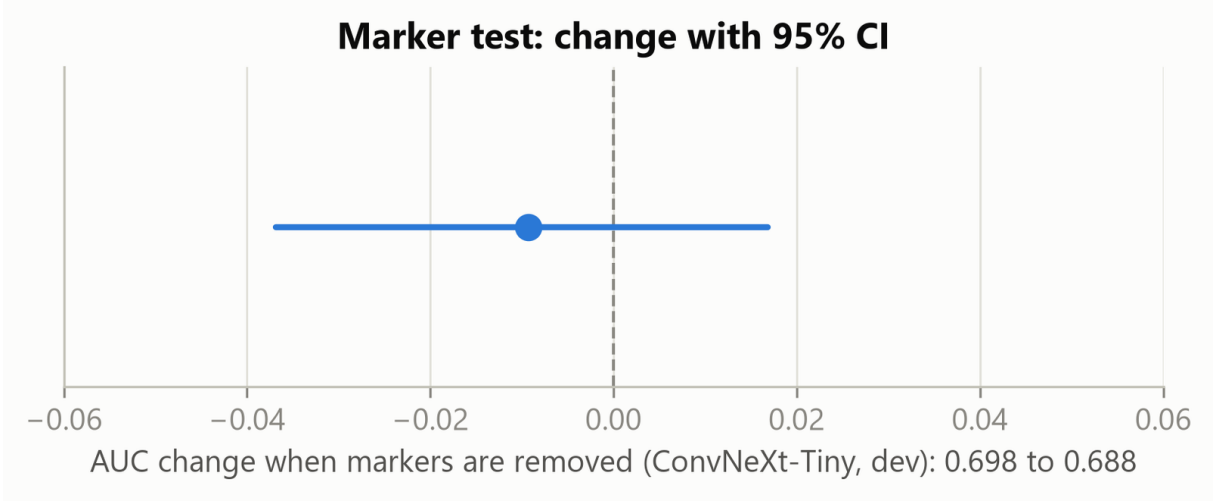


Figure 21. *Marker test: AUC with and without markers removed by inpainting (ConvNeXt-Tiny on Task A dev).*

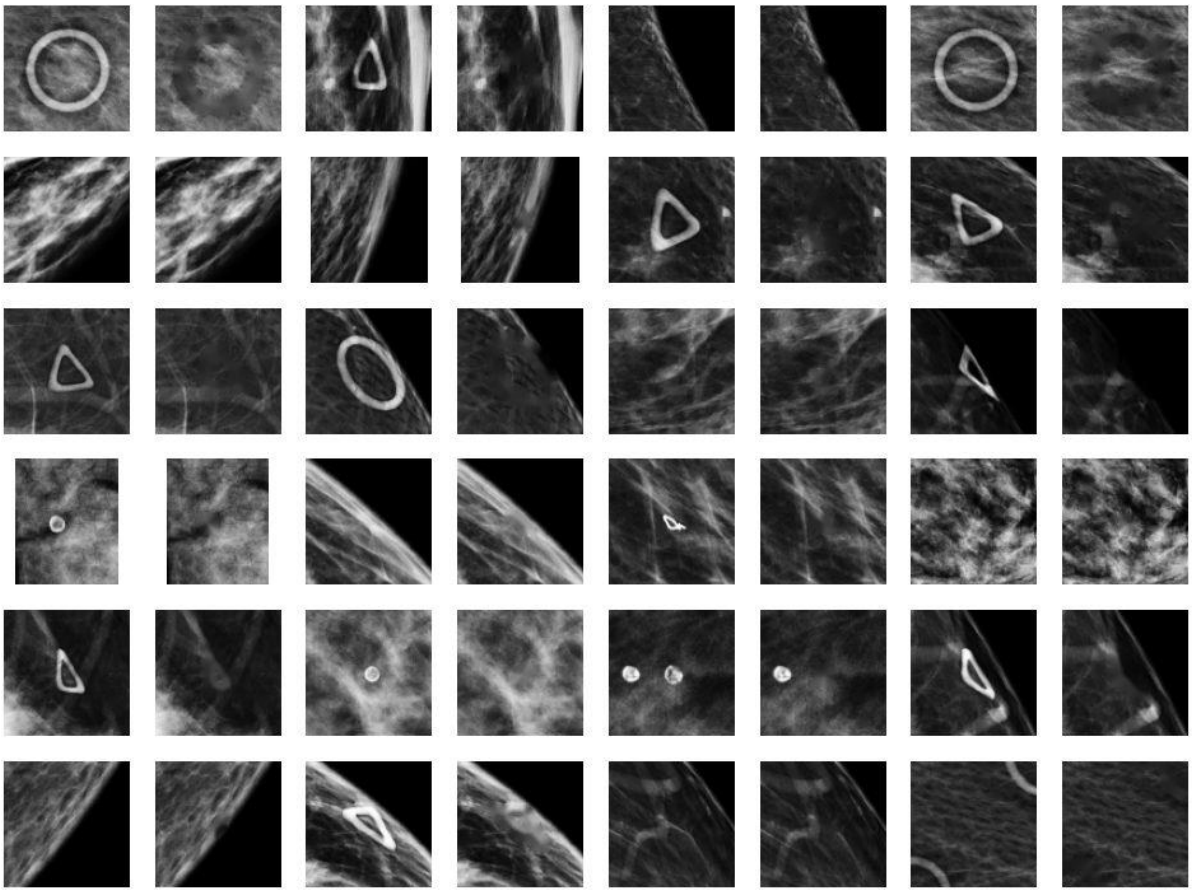


Figure 22. *Mask zoom: an example of marker detection and inpainting. The detector catches ~8% of images, erases some rim calcifications and misses some solid dots.*

So **no clear reliance on markers** was found for Task A, but a small effect is not excluded. The test covers one model on dev, and the detector has the limits described in Figure 22. BI-RADS was *not* re-tested under masking.

5. Limitations

The 90% target was not reached. The main limitations are:

1. **Every Task A exam led to a biopsy.** The benign cases already looked suspicious to a radiologist. The task is therefore closer to "which biopsy-indicated exam turns out malignant" than to screening. **Nothing in this report applies to screening performance.**
2. **Single institution.** EMBED comes from one institution (four Emory hospitals). There is no external validation. Image appearance, scanner mix, population and reporting style will all shift elsewhere.

3. **Test–dev gap.** Test AUCs run above dev AUCs. Part of this is because each test exam is scored by the mean of 5 fold models, while each dev exam is scored by the one fold model that did not train on it. Confidence intervals are wide (± ~0.05 on Task A AUC).
4. **The combined model includes the 768×512 baseline**, which was trained before the crop fix in v2. The result is still the honest headline number — a reader who prefers a clean cut can take the Kaggle v2 stack alone (image-only test AUC 0.767).
5. **BI-RADS "image performance" may largely reflect an exam-type shortcut**, as discussed in §4.2. Reading-skill claims should not be made from the Task B numbers in this form. Each class is also capped at 2,500 exams, so these scores are not on the natural class mix.
6. **The headline leans on the radiologist's own reading.** Test AUC rises from 0.776 (images) to 0.799 (+ exam type, age) to 0.826 (+ findings), with overlapping CIs. On dev, in the baseline run, the radiologist's reading (including the BI-RADS assessment) with exam type and age and no images reaches AUC 0.797; adding the images reaches 0.809 (`outputs/metrics/a_clinical_field_trial.json`). So on dev the images add about 0.01 beyond the reading. This comparison was not run on test.
7. **The test set was opened once.** Any further model compared on the same test exams must say so explicitly.
8. **Small sample.** Task A has 1,757 breast exams from 1,513 patients, only 517 of them with confirmed cancer, and the open release is 20% of the full 2D and C-view dataset.

6. Reproducibility and artifacts

- **Metrics (ground truth for all numbers in this report):** `outputs_v2/metrics/final_summary.csv`, `a_combined.json`, `a_metrics.json`, `b_metrics.json`, `b_metadata_baseline.json`, `a_test_subgroups.csv`, `outputs/metrics/a_clinical_field_trial.json`, `outputs_masked/metrics/masked_vs_baseline.json`.
- **Figures:** `outputs_v2/figures/final/` (8 summary plots), `outputs_v2/figures/a/` (24), `outputs_v2/figures/b/` (20), `outputs/figures/a/` (24 baseline).
- **Proof pack:** `outputs_v2/proof/manifest.csv` (file hashes), `kaggle_env.json`, `kaggle_pip_freeze.txt`, `kaggle_nvidia_smi.txt`, `test_plan_malignant.md`.
- **Code:** `scripts/s0…s7`, `scripts/run_pipeline.py`, `scripts/combine_runs.py`, `kaggle/{make_jobs.py, runner.py, fetch_output.py, import_results.py}`.
- **Settings:** `configs/kaggle_v2.json`, `configs/full_run.json`.
- **Checkpoints:** `best.pt` for every model on the PC; `last.pt` for 14 of the 15 Task A Kaggle runs is only on Kaggle (`kaggle/fetch_output.py --all`).
- **Environment:** Python 3.11, `.venv/`.
- **Error history (what broke and how it was fixed):** `docs/fix_log.md`.

To re-run one pipeline and open its test set:

```
EMBED_RUN=outputs_v2 EMBED_CONFIG=kaggle_v2.json \
python scripts/run_pipeline.py --final
```

The headline combination needs both the `outputs/` and `outputs_v2/` results, then `python scripts/combine_runs.py`. The `outputs_v2/` training ran on Kaggle through `kaggle/`.

7. Conclusions and next steps

What this run established.

- On biopsy-indicated EMBED exams, a six-model image combination reaches **AUC 0.776** on an unseen test set; adding exam type and age brings it to **0.799**; adding the radiologist's pre-biopsy findings brings it to **0.826**. On dev, the radiologist's reading with exam type and age, without images, already reaches 0.797.
- BI-RADS classification from images on EMBED may be dominated by an exam-type shortcut: exam type alone nearly matches the image stack on dev, so these scores are not a fair measure of reading skill.
- No clear evidence that the Task A model relies on image markers, from a limited test (one model, dev set); a small effect is not excluded.
- **Overall accuracy does not reach the 90% target** the team originally set. The model is **not fit for clinical use** and the result does **not** apply to screening.

Candidate next directions (a decision for the team, not this report):

1. **Cancer vs. normal with lesion localization on EMBED.** EMBED ships bounding boxes on 9,693 standard-view images from 4,503 patients; 924 of these overlap Task A. Boxes would need to be mapped through the crop/flip/resize pipeline (no pixel outlines are provided).
2. **More confirmed-cancer cases from other datasets.** EMBED's open table has only 658 patients with confirmed cancer, and Task A uses 517 of them.
3. **A new problem domain** — the team has discussed brain-bleed CT (CQ500 remains available), diabetic retinopathy or fractures.

The `data/embed/png` folder (3.1 GB) is kept until a decision on direction (1) is made, and can be deleted otherwise.

References

[1] Jeong JJ, Vey BL, Bhimireddy A, et al. The EMory BrEast imaging Dataset (EMBED): A Racially Diverse, Granular Dataset of 3.4 Million Screening and Diagnostic Mammographic Images. Radiology: Artificial Intelligence. 2023;5(1):e220047. doi:10.1148/ryai.220047

[2] He K, Zhang X, Ren S, Sun J. Deep Residual Learning for Image Recognition. CVPR 2016.

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[4] Liu Z, Mao H, Wu CY, Feichtenhofer C, Darrell T, Xie S. A ConvNet for the 2020s. CVPR 2022.

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Appendix A — Proof of work

Log panels rendered from the saved logs of the actual runs on Kaggle and the local PC. Code was run in this project folder and on Kaggle. Checkpoints and output files are listed in §6.

A.1 Kaggle GPU and training logs

```
nvidia-smi on the Kaggle machine
Rendered from saved logs on 2026-11-01T13:36:39

Sat Oct 31 19:44:50 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 580.178.04             Driver Version: 580.178.04     CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  Tesla T4                       Off |   00000000:00:04.0 Off |                    0 |
| N/A   36C    P8             12W /   70W |       3MiB /  15360MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+
|   1  Tesla T4                       Off |   00000000:00:05.0 Off |                    0 |
| N/A   38C    P8             11W /   70W |       3MiB /  15360MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
```

Figure A1. *Kaggle GPU allocation (two Tesla T4) recorded at the start of the Task A job.*

```
Kaggle job embed-malignant: stages
Rendered from saved logs on 2026-11-01T13:36:37

[2026-10-31T19:44:52] [kaggle] KAGGLE_START flag=in_task_a runs=15
[2026-10-31T20:04:40] [kaggle] STAGE_DONE fetch ok=3407 failed=0 in 19.6 min
[2026-10-31T21:15:09] [kaggle] STAGE_DONE a_efficientnet_v2_s_fold1 on gpu1 in 70.5 min
[2026-10-31T21:27:10] [kaggle] STAGE_DONE a_efficientnet_v2_s_fold0 on gpu0 in 82.5 min
[2026-10-31T22:20:35] [kaggle] STAGE_DONE a_efficientnet_v2_s_fold2 on gpu1 in 65.4 min
[2026-10-31T22:27:37] [kaggle] STAGE_DONE a_efficientnet_v2_s_fold3 on gpu0 in 60.5 min
[2026-10-31T23:20:35] [kaggle] STAGE_DONE a_efficientnet_v2_s_fold4 on gpu1 in 60.0 min
[2026-10-31T23:44:38] [kaggle] STAGE_DONE a_convnext_tiny_fold0 on gpu0 in 77.0 min
[2026-11-01T00:47:44] [kaggle] STAGE_DONE a_convnext_tiny_fold1 on gpu1 in 87.2 min
[2026-11-01T01:01:13] [kaggle] STAGE_DONE a_convnext_tiny_fold2 on gpu0 in 76.6 min
[2026-11-01T02:09:06] [kaggle] STAGE_DONE a_convnext_tiny_fold3 on gpu1 in 81.4 min
[2026-11-01T02:17:37] [kaggle] STAGE_DONE a_convnext_tiny_fold4 on gpu0 in 76.4 min
[2026-11-01T03:19:57] [kaggle] STAGE_DONE a_resnet50_fold0 on gpu1 in 70.8 min
[2026-11-01T03:28:37] [kaggle] STAGE_DONE a_resnet50_fold1 on gpu0 in 71.0 min
[2026-11-01T03:48:56] [kaggle] STAGE_DONE a_resnet50_fold2 on gpu1 in 29.0 min
[2026-11-01T04:22:24] [kaggle] STAGE_DONE a_resnet50_fold4 on gpu1 in 33.5 min
[2026-11-01T04:35:01] [kaggle] STAGE_DONE a_resnet50_fold3 on gpu0 in 66.4 min
[2026-11-01T04:35:57] [kaggle] KAGGLE_DONE
```

Figure A2. *Kaggle job log — Task A (malignant vs. benign), all 15 runs.*

```
Kaggle job embed-birads: stages
Rendered from saved logs on 2026-11-01T13:36:38

[2026-10-31T20:05:28] [kaggle] KAGGLE_START flag=in_task_b runs=3
[2026-10-31T20:44:09] [kaggle] STAGE_DONE fetch ok=22086 failed=0 in 38.4 min
[2026-10-31T22:26:44] [kaggle] STAGE_DONE b_efficientnet_v2_s_fold0 on gpu0 in 102.6 min
[2026-10-31T23:13:46] [kaggle] STAGE_DONE b_convnext_tiny_fold0 on gpu1 in 149.6 min
[2026-11-01T00:35:09] [kaggle] STAGE_DONE b_resnet50_fold0 on gpu0 in 128.4 min
[2026-11-01T00:36:28] [kaggle] KAGGLE_DONE
```

Figure A3. *Kaggle job log — Task B (BI-RADS), all 3 runs.*

A.2 Local PC evaluation

```
Second-pass evaluation on the PC
Rendered from saved logs on 2026-11-01T13:36:38

[2026-11-01T09:28:17] [evaluate] STAGE_DONE evaluate b unlock_test=False
[2026-11-01T09:36:04] [figures] STAGE_DONE figures b: 11 figures
[2026-11-01T09:37:58] [evaluate] b stacker candidates on dev: {'mean_of_models': 0.3947863566253103, 'logistic': 0.3988877561782735, 'lightgbm': 0.396
[2026-11-01T09:39:46] [evaluate] b dev image stack 0.3989, with exam type and age 0.4892
[2026-11-01T09:39:46] [evaluate] b dev stack: acc=0.4266 bal_acc=0.3989 macro_f1=0.4080 macro_auc=0.7587981513892247
[2026-11-01T09:39:46] [evaluate] STAGE_DONE evaluate b unlock_test=False
[2026-11-01T09:40:53] [figures] STAGE_DONE figures b: 12 figures
[2026-11-01T09:43:16] [evaluate] b stacker candidates on dev: {'mean_of_models': 0.3947863566253103, 'logistic': 0.3988877561782735, 'lightgbm': 0.396
[2026-11-01T09:45:05] [evaluate] b dev image stack 0.3989, with exam type and age 0.4892
[2026-11-01T09:46:57] [evaluate] b test stack: acc=0.4361 bal_acc=0.3911 macro_f1=0.4097 macro_auc=0.7807175356870598
[2026-11-01T09:46:57] [evaluate] STAGE_DONE evaluate b unlock_test=True
[2026-11-01T09:50:44] [figures] STAGE_DONE figures b: 18 figures
[2026-11-01T13:14:28] [evaluate] a stacker candidates on dev: {'mean_of_models': 0.723414256293327, 'logistic': 0.7226109262085578, 'lightgbm': 0.7055
[2026-11-01T13:16:05] [evaluate] a dev image stack 0.7234, with exam type and age 0.7474
[2026-11-01T13:16:23] [evaluate] a dev with radiologist reading, exam type and age 0.8122
[2026-11-01T13:16:36] [evaluate] a dev stack: AUC=0.7234 acc@maxacc=0.7218 sens=0.247 spec=0.950
[2026-11-01T13:16:36] [evaluate] STAGE_DONE evaluate a unlock_test=False
[2026-11-01T13:20:42] [figures] STAGE_DONE figures a: 14 figures
[2026-11-01T13:21:21] [evaluate] a stacker candidates on dev: {'mean_of_models': 0.723414256293327, 'logistic': 0.7226109262085578, 'lightgbm': 0.7055
[2026-11-01T13:22:40] [evaluate] a dev image stack 0.7234, with exam type and age 0.7474
[2026-11-01T13:22:55] [evaluate] a dev with radiologist reading, exam type and age 0.8122
[2026-11-01T13:24:41] [evaluate] a test stack: AUC=0.7669 acc@maxacc=0.7213 sens=0.223 spec=0.958
[2026-11-01T13:24:41] [evaluate] STAGE_DONE evaluate a unlock_test=True
[2026-11-01T13:34:54] [figures] STAGE_DONE figures a: 22 figures
```

Figure A4. *PC evaluation log — stacker candidates and dev and test scores for the second-pass run.*